Sufficient and Necessary Conditions for the Parameters of Conditional Entropies

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Abstract

This standalone note gives the sufficient and necessary parameter conditions for the conditional entropies introduced in Ref. [4]. It is a concise, human-readable replacement for Sections 4 and 5 of that paper. The statements cover finite and general measures, both tropical endpoints, and derivations. In particular, no global finiteness assumption is imposed on the measure. The proofs retain the main ideas—finite-support factorization, common discretization, and localized high-dimensional counterexamples—while omitting the lengthy analytic calculations.

Contents

1 Sufficient conditions for the parameters

In this section, we establish sufficient conditions for the parameters under which the homomorphisms and derivations (and, consequently, the associated conditional entropies) are monotone under conditional majorization. In the following section, we show that these conditions are also necessary, without imposing a separate regularity assumption on the measure.

In Ref. [4], the general forms of the homomorphisms and derivations depend on a finite measure: in the temperate case one may write it as $\mu = t\tau$, with τ a probability measure and its sign encoded by t. However, not all such choices are valid, as some lead to violations of monotonicity under conditionally mixing channels. While monotonicity under mixing channels acting on the first register always holds, monotonicity under channels acting on the conditioning register fails for some choices of the parameters. We therefore examine the convexity properties of the homomorphism as a function of the joint probability matrix. Proposition A.1 and Lemma A.3 of Ref. [4] identify these convexity properties with the required monotonicity and lift the columnwise result to the original semiring and channel preorder of Eqs. (43) and (49) therein.

Let us first identify the function for which we need to establish convexity or concavity properties. In the temperate case, Eq. (78) and Proposition 3.6 of Ref. [4] give

$$\Phi(P_{XY}) = \Phi_{t,\tau}(P_{XY}) = \sum_{y\in\mathcal{Y}} P_Y(y) \exp\left(t \int_{[0,+\infty]} H_\alpha(P_{X|Y=y})\ \mathrm{d}\tau(\alpha) \right), \tag{1}$$

where τ is a probability measure and $t \in \mathbb{R}\setminus\{0\}$. Thus it suffices to study, for $\vec{x} \in \mathbb{R}^d_{\geq 0}\setminus\{0\}$,

$$\vec{x} \longmapsto \|\vec{x}\|_1 \exp\left(t \int_{[0,+\infty]} H_\alpha(\hat{x})\ \mathrm{d}\tau(\alpha) \right), \qquad \hat{x} = \frac{\vec{x}}{\|\vec{x}\|_1}. \tag{2}$$

The Rényi family and its endpoint values H_0, H_1, and H_∞ are those of Eq. (2) and the sentence following it in Ref. [4]. We begin with finite support, pass to a general measure by a common discretization, and finally take the tropical and derivation limits.

1.1 Sufficient conditions for the discrete case

The discrete case exploits two basic facts. The first is Minkowski's inequality [1, Theorem 9].

Lemma 1.1. *Let $\alpha \in \mathbb{R}\setminus\{0\}$. The function, defined on $\mathbb{R}^d_{>0}$,*

$$\vec{x} \longmapsto \left(\sum_{i=1}^d x_i^\alpha \right)^{1/\alpha}, \tag{3}$$

1. *is convex if $\alpha > 1$;*
2. *is concave if $\alpha < 1$.*

The displayed power mean is not defined at $\alpha = 0$. Whenever the entropy parameter equals 0, the proof instead uses the endpoint formula for H_0.

The second fact is the standard convexity criterion for multivariate monomials [2, Lemma 8]; see also Ref. [3, Proposition 13].

Lemma 1.2. *Let $N \in \mathbb{N}$ and let $\vec{\beta} = (\beta_1, \ldots, \beta_N) \in \mathbb{R}^N$ satisfy $\sum_{\ell=1}^N \beta_\ell = 1$. The function, defined on $\mathbb{R}^N_{>0}$,*

$$\vec{y} \longmapsto \prod_{\ell=1}^N y_\ell^{\beta_\ell}, \tag{4}$$

1. *is concave if and only if $\beta_\ell \geq 0$ for every ℓ;*
2. *is convex if and only if there is a unique index k such that $\beta_k > 0$ and $\beta_\ell \leq 0$ for every $\ell \neq k$.*

Combining these facts gives the finite-support result.

Lemma 1.3. *Let τ be a discrete probability measure supported on finitely many points $\alpha_1 < \cdots < \alpha_N$ of $[0, +\infty]$, and let $t \in \mathbb{R}\setminus\{0\}$. For every $d \in \mathbb{N}$, the function*

$$\vec{x} \longmapsto \|\vec{x}\|_1 \exp\left(t \int_{[0,+\infty]} H_\alpha(\hat{x})\ \mathrm{d}\tau(\alpha) \right) \tag{5}$$

on $\mathbb{R}^d_{\geq 0}\setminus\{0\}$ is

1. *concave if* $t > 0$, $\alpha_1, \ldots, \alpha_N < 1$, *and*

$$\sum_{\ell=1}^{N} \frac{\alpha_\ell}{1-\alpha_\ell} \tau(\{\alpha_\ell\}) \leq \frac{1}{t}; \tag{6}$$

2. *convex if* $t < 0$ *and either*

   (a) $\alpha_1, \ldots, \alpha_N \leq 1$, *or*

   (b) $\alpha_1, \ldots, \alpha_{N-1} < 1$, $\alpha_N > 1$, *and*

$$\sum_{\ell=1}^{N} \frac{\alpha_\ell}{1-\alpha_\ell} \tau(\{\alpha_\ell\}) \leq \frac{1}{t}, \tag{7}$$

   where $\alpha/(1-\alpha)$ *has value* -1 *at* $\alpha = +\infty$.

Proof idea. First assume that every atom satisfies $0 < \alpha_\ell < +\infty$ and $\alpha_\ell \neq 1$, and set

$$\beta_\ell = t \frac{\alpha_\ell}{1-\alpha_\ell} \tau(\{\alpha_\ell\}), \qquad \bar{\beta} = 1 - \sum_{\ell=1}^{N} \beta_\ell. \tag{8}$$

The function factors as

$$\left(\sum_i x_i \right)^{\bar{\beta}} \prod_{\ell=1}^{N} \left(\sum_i x_i^{\alpha_\ell} \right)^{\beta_\ell/\alpha_\ell}. \tag{9}$$

Lemma 1.1 supplies the convexity or concavity of each power mean, and Lemma 1.2 supplies the joint sign criterion for the outer monomial. When $t < 0$ and all support lies in $[0, 1]$, no moment bound is needed. Atoms at 0, 1, and $+\infty$ are handled by the endpoint formulas and endpoint-preserving perturbations or pointwise limits; the branch containing $\alpha = 1$ occurs only in the all-lower negative case. ☐

1.2 Sufficient conditions for the temperate homomorphisms - general measure

We now extend the result to a general probability measure τ. The signed integral

$$\int_{[0,1)} \frac{\alpha}{1-\alpha} \, \mathrm{d}\tau(\alpha) + \int_{(1,+\infty]} \frac{\alpha}{1-\alpha} \, \mathrm{d}\tau(\alpha) \leq \frac{1}{t} \tag{10}$$

is used only when the two one-sided integrals are finite; at $+\infty$ the integrand is -1. In particular, Eq. (10) never hides an undefined cancellation at $\alpha = 1$.

Proposition 1.4. *Let* τ *be a probability measure on* $[0, +\infty]$ *and let* $t \in \mathbb{R} \setminus \{0\}$. *The function in Eq.* (2) *is*

1. *concave if* $t > 0$, $\mathrm{supp}(\tau) \subseteq [0, 1]$, $\tau(\{1\}) = 0$, *and*

$$\int_{[0,1)} \frac{\alpha}{1-\alpha} \, \mathrm{d}\tau(\alpha) \leq \frac{1}{t}; \tag{11}$$

2. *convex if* $t < 0$ *and either*

   (a) $\mathrm{supp}(\tau) \subseteq [0, 1]$, *with no moment-finiteness condition, or* *(b)* *there is* $\alpha_* \in (1, +\infty]$ *such that* $\operatorname{supp}(\tau) \subseteq [0,1] \cup \{\alpha_*\}$, $\tau(\{1\}) = 0$, *the two one-sided integrals in Eq.* (10) *are finite, and Eq.* (10) *holds.*

Proof idea. Use one partition of $[0, +\infty]$ for every vector appearing in the convexity inequality and push the mass of each cell to a common representative. On $[0, 1)$ the representatives are chosen from below, so the discrete moment bound is preserved. The Shannon branch is kept separate, and in the exceptional negative case the point α_* is fixed throughout the approximation. Each finite approximation satisfies Lemma 1.3.

For fixed dimension, $0 \le H_\alpha(\hat{x}) \le \log d$, including the endpoint values. Dominated convergence therefore gives pointwise convergence of the entropy integrals and of their exponentials, and a pointwise limit preserves the required convexity or concavity inequality. This is the only convergence input needed here. Notice especially that the all-lower branch with $t < 0$ requires no integrability of $\alpha/(1-\alpha)$ near 1. □

1.3 Monotonicity under mixing channels

As mentioned at the beginning of this section, monotonicity under mixing channels acting on the first register X always holds. Mixing on X increases every $H_\alpha(P_X)$ and hence the integrated entropy. Consequently $\Phi_{t,\tau}$ increases for $t > 0$ and decreases for $t < 0$; after applying $(1/t)\log$, the associated conditional entropy increases in both cases. This gives the appropriate positive- and negative-homomorphism order directions. Proposition A.1 and Lemma A.3 of Ref. [4] then convert the columnwise concavity or convexity established above into monotonicity under the conditioning operation and lift the two components to the original conditionally mixing channels. For derivations, the corresponding statements are Lemmas A.2 and A.4 of that reference.

1.4 Sufficient conditions for tropical homomorphisms

We next take the tropical limits. Monotonicity is preserved under pointwise limits, so the temperate conditions immediately provide the admissible endpoint families.

Proposition 1.5. *Let* τ *be a probability measure on* $[0, +\infty]$*. The function*

$$P_{XY} \longmapsto \min_{\substack{y \in \mathcal{Y} \\ P_Y(y) > 0}} \int_{[0,+\infty]} H_\alpha(P_{X|Y=y})\, \mathrm{d}\tau(\alpha) \tag{12}$$

is monotone under conditionally mixing channels if either

(a) $\operatorname{supp}(\tau) \subseteq [0,1]$, *with no moment-finiteness condition, or*

(b) *there is* $\alpha_* \in (1, +\infty]$ *such that* $\operatorname{supp}(\tau) \subseteq [0,1] \cup \{\alpha_*\}$, $\tau(\{1\}) = 0$, *both one-sided integrals are finite, and*

$$\int_{[0,1)} \frac{\alpha}{1-\alpha}\, \mathrm{d}\tau(\alpha) + \int_{(1,+\infty]} \frac{\alpha}{1-\alpha}\, \mathrm{d}\tau(\alpha) \le 0. \tag{13}$$

Proof idea. Use

$$\lim_{t \to -\infty} \frac{1}{t} \log\left(\sum_{y \in \mathcal{Y}} P_Y(y) \exp\left(t \int H_\alpha(P_{X|Y=y})\, \mathrm{d}\tau(\alpha) \right) \right) \tag{14}$$

$$= \min_{\substack{y \in \mathcal{Y} \\ P_Y(y) > 0}} \int H_\alpha(P_{X|Y=y})\, \mathrm{d}\tau(\alpha). \tag{15}$$

The all-lower case is the limit of Proposition 1.4(2a). If the signed moment is strictly negative, the exceptional case holds for all sufficiently large $|t|$. At equality, move an arbitrarily small amount of mass to α_*, apply the strict case, and take a final pointwise limit. This retains both normalization and the boundary of the parameter range. $\square$

Proposition 1.6. *The function*

$$P_{XY} \longmapsto \max_{\substack{y\in\mathcal{Y} \\ P_Y(y)>0}} H_0(P_{X|Y=y}) \tag{16}$$

is monotone under conditionally mixing channels.

Proof idea. This is the $t \to +\infty$ limit of the positive temperate branch with τ concentrated at $\alpha = 0$:

$$\lim_{t\to+\infty} \frac{1}{t} \log\left(\sum_{y\in\mathcal{Y}} P_Y(y) e^{tH_0(P_{X|Y=y})}\right) = \max_{\substack{y\in\mathcal{Y} \\ P_Y(y)>0}} H_0(P_{X|Y=y}). \tag{17}$$

$\square$

Remark 1.7. Unless $\alpha = 0$, the positive temperate moment bound is eventually violated as $t \to +\infty$. Thus the positive tropical family contains only the $\alpha = 0$ point mass. In contrast, every point mass is allowed in the negative tropical family, subject to the two alternatives in Proposition 1.5.

1.5 Sufficient conditions for derivations

We now turn to the derivations.

Proposition 1.8. *The function*

$$P_{XY} \longmapsto \sum_{y\in\mathcal{Y}} P_Y(y) H_\alpha(P_{X|Y=y}) \tag{18}$$

is monotone under conditionally mixing channels if $\alpha \in [0, 1]$.

Proof idea. For the point mass at the actual parameter α, take $t \to 0^-$ in the all-lower branch of Proposition 1.4. The finite log-sum-exp identity

$$\lim_{t\to0} \frac{1}{t} \log\left(\sum_{y\in\mathcal{Y}} P_Y(y) e^{tH_\alpha(P_{X|Y=y})}\right) = \sum_{y\in\mathcal{Y}} P_Y(y) H_\alpha(P_{X|Y=y}) \tag{19}$$

passes monotonicity to the limit. Equivalently, this is the concavity of the perspective $\vec{x} \mapsto \|\vec{x}\|_1 H_\alpha(\hat{x})$ used in Lemma A.2 of Ref. [4]. $\square$

These results identify the admissible homomorphisms and derivations. With the normalizations of Definition 6.1 in Ref. [4], they give the families in Eqs. (122)–(125) of that reference.

2 Necessary conditions for the parameters

In this section, we show that the sufficient conditions on the parameters derived in the previous section are also necessary. No assumption is made that the measure vanishes at a prescribed rate as $\alpha \to 1$, and no global finiteness condition is placed on the weighted integral. In the positive branch, concavity forces the lower one-sided moment to be finite and bounded. In the exceptional negative branch, convexity forces both one-sided moments to be finite, whereas the all-lower negative branch requires no such finiteness.

Our approach consists in identifying explicit perturbation directions and computing the signs of the second derivatives along them. Outside the ranges of Section 1, the resulting curvature has the wrong sign. Sharp counterexamples require increasingly high dimensions, but the underlying idea is the finite-dimensional multivariate monomial

$$\vec{v} \longmapsto v_1^{\beta_0} v_2^{\beta_1} \cdots v_d^{\beta_d}, \qquad \beta_0 = 1 - \sum_{i=1}^{d} \beta_i, \tag{20}$$

whose curvature criterion is Lemma 1.2. In the discrete entropy expression, the correspondence is

$$\beta_i \longleftrightarrow \tau(\{\alpha_i\}) t \frac{\alpha_i}{1-\alpha_i}, \qquad v_i \longleftrightarrow \left(\sum_j x_j^{\alpha_i} \right)^{1/\alpha_i}. \tag{21}$$

The latter quantities are coupled through $\vec{x}$. Blocks whose cardinalities and weights scale with a dimension parameter make different blocks dominate on different ranges of α. For example, appending d entries of weight $1/d$ gives

$$\left(\sum_j y_j^{\alpha} \right)^{1/\alpha} = \left(\sum_{j=1}^{k} x_j^{\alpha} + d^{1-\alpha} \right)^{1/\alpha}, \tag{22}$$

which diverges for $\alpha < 1$ and converges to $(\sum_{j=1}^{k} x_j^{\alpha})^{1/\alpha}$ for $\alpha > 1$.

There is one analytic point that is important even in this abbreviated account. For each localized path used below, its support is fixed near the perturbation point and one maximizing coordinate remains fixed there. On these paths the endpoint-aware kernels and their first two path derivatives extend continuously through 0, 1, and $+\infty$; in particular, the apparent pole at 1 is removable. One differentiates the finite-dimensional kernels before integrating, then truncates the high-dimensional analysis on the two sides of 1 and passes to the endpoint only afterward. Domination by total variation and monotone convergence make the required moment bounds conclusions of the argument. Thus no derivative is exchanged with an already limiting piecewise envelope, and no separate finiteness assumption is needed.

2.1 Necessary conditions; temperate case

The temperate case corresponds to the bulk entropies $H_{t,\tau}$ in Eq. (122) of Ref. [4]. As above, it is sufficient to study Eq. (2). For notational convenience, combine the weight t with the probability measure τ and write

$$\mu := t\tau. \tag{23}$$

Whenever the two one-sided integrals are finite, we also have the coefficient

$$\beta_0 = 1 - \int_{[0,+\infty]\setminus\{1\}} \frac{\alpha}{1-\alpha} \, \mathrm{d}\mu(\alpha), \tag{24}$$

where the integral is the sum of its parts on $[0, 1)$ and $(1, +\infty]$.

Whenever a signed moment is written as a single ordinary real number below, it is the sum of the two one-sided integrals after their finiteness has been established. Finiteness is not a blanket hypothesis on μ: before it is known, the argument works with endpoint-separated truncations. The localized estimates show that if convexity leaves any support above 1, the relevant one-sided moments must be finite. If all support lies in $[0, 1]$, no such condition is imposed.

We first treat $\mu > 0$, and then $\mu < 0$. In the negative case the Shannon atom, the signed moment, and the number of upper support points are isolated in separate steps, exactly matching the alternatives in Proposition 1.4.

Proposition 2.1. *Let μ be a positive finite measure on $[0, +\infty]$. The function*

$$\vec{x} \longmapsto \|\vec{x}\|_1 \exp\left(\int_{[0,+\infty]} H_\alpha(\hat{x}) \, \mathrm{d}\mu(\alpha) \right) \tag{25}$$

is concave only if $\operatorname{supp}(\mu) \subseteq [0, 1]$, $\mu(\{1\}) = 0$, *and*

$$\int_{[0,1)} \frac{\alpha}{1 - \alpha} \, \mathrm{d}\mu(\alpha) \leq 1. \tag{26}$$

Proof idea. For $p \in (0, 1)$, use the two-block point and traceless direction

$$\vec{x} = \Bigg[\underbrace{\frac{p}{d}, \ldots, \frac{p}{d}}_{d \text{ times}}, \underbrace{\frac{1-p}{d^2}, \ldots, \frac{1-p}{d^2}}_{d^2 \text{ times}} \Bigg], \tag{27}$$

$$\vec{v} = \Bigg[\underbrace{-\frac{1}{d}, \ldots, -\frac{1}{d}}_{d \text{ times}}, \underbrace{\frac{1}{d^2}, \ldots, \frac{1}{d^2}}_{d^2 \text{ times}} \Bigg]. \tag{28}$$

As $d \to \infty$, the two blocks localize the ranges below and above 1. Any Shannon mass produces a strictly positive leading curvature. Upper mass gives a coefficient $\beta < 0$, hence $\beta^2 - \beta > 0$, and a lower moment exceeding 1 gives the same wrong sign with $\beta > 1$. Each case contradicts concavity. The truncation procedure described above justifies the limits and simultaneously shows that the lower moment is finite and bounded by 1. ☐

Next, let μ be negative.

Proposition 2.2. *Let μ be a negative finite measure with $\mu(\{1\}) \neq 0$. Then the function*

$$\vec{x} \longmapsto \|\vec{x}\|_1 \exp\left(\int_{[0,+\infty]} H_\alpha(\hat{x}) \, \mathrm{d}\mu(\alpha) \right) \tag{29}$$

is convex only if $\operatorname{supp}(\mu) \subseteq [0, 1]$.

Proof idea. Assume that both the Shannon atom and upper support are present. If $\mu(\{1\}) < 0$ and $\beta_1 > 0$ is a localized upper contribution, the two-block direction is scaled as $(-\beta_1/\mu(\{1\}), 1)$. The first logarithmic derivative then tends to zero and the second tends to $-\beta_1 < 0$. This is a genuinely concave direction and contradicts convexity. Working first on truncated ranges makes the calculation finite; the endpoint limits are taken only after the curvature sign is fixed. ☐

Proposition 2.3. *Let μ be a negative finite measure with $\mu(\{1\}) = 0$, and suppose that the function*

$$\vec{x} \longmapsto \|\vec{x}\|_1 \exp\left(\int_{[0,+\infty]} H_\alpha(\hat{x})\ \mathrm{d}\mu(\alpha)\right) \tag{30}$$

is convex. If $\operatorname{supp}(\mu)$ *meets* $(1, +\infty]$*, then both one-sided integrals are finite and*

$$\int_{[0,1)} \frac{\alpha}{1-\alpha}\ \mathrm{d}\mu(\alpha) + \int_{(1,+\infty]} \frac{\alpha}{1-\alpha}\ \mathrm{d}\mu(\alpha) \geq 1. \tag{31}$$

Consequently, a finite signed sum smaller than 1 *is compatible with convexity only if* $\operatorname{supp}(\mu) \subseteq [0, 1]$.

Proof idea. First use separated upper neighborhoods with stationary block velocities. If two distinct upper portions survived, their limiting second derivative would be negative, so convexity leaves at most one upper point; its upper moment is therefore finite. A three-block construction then controls normalization and localizes the lower and upper ranges. The perturbation is chosen so that the first derivative is exactly zero. On every truncation away from 1, a signed moment smaller than 1 makes the limiting second derivative negative. Convexity rules this out. Monotone passage through the truncations forces finiteness of the lower moment and gives Eq. (31). This is how endpoint localization replaces the old global finiteness hypothesis. □

Remark 2.4. The points used in these counterexamples need not initially be normalized. The function is positively homogeneous on the whole semiring domain, so rescaling by a positive constant does not change the curvature sign. Equivalently, the counterexample can be embedded into a normalized joint probability matrix and then interpreted in the semiring of Section 3B and lifted using Proposition A.1 and Lemma A.3 of Ref. [4].

Remark 2.5. A nontraceless direction can likewise be embedded into a trace-preserving two-column perturbation. If the first column changes as $\vec{x} + \lambda\vec{v}$, change the second as

$$\vec{x}' - \lambda\hat{x}' \sum_i v_i. \tag{32}$$

The opposite sign is essential: it keeps the total weight fixed. Additivity over the conditioning register makes the second column affine in λ, so it does not alter the curvature obstruction.

Finally, the upper support cannot contain two distinct points.

Proposition 2.6. *Let μ be a negative finite measure with $\mu(\{1\}) = 0$, and suppose that the function*

$$\vec{x} \longmapsto \|\vec{x}\|_1 \exp\left(\int_{[0,+\infty]} H_\alpha(\hat{x})\ \mathrm{d}\mu(\alpha)\right) \tag{33}$$

is convex. Then either $\operatorname{supp}(\mu) \subseteq [0, 1]$*, or there is a unique* $\alpha_* \in (1, +\infty]$ *such that*

$$\operatorname{supp}(\mu) \subseteq [0, 1] \cup \{\alpha_*\}. \tag{34}$$

In the second case the two one-sided moments are finite and satisfy Eq. (31).

Proof idea. If two disjoint portions of the support lie above 1, localize them in two different dominant blocks. Their positive localized weights may be denoted $\beta_1, \beta_2 > 0$. The block velocities are chosen to cancel the first derivative, while the limiting second derivative is

$$-\frac{1}{\beta_1} - \frac{1}{\beta_2} < 0. \tag{35}$$

This contradicts convexity. Shrinking the two localization windows shows that the upper support has at most one point. The finiteness and signed inequality in the exceptional branch are the conclusion of the joint truncation argument in Proposition 2.3. □

Thus the last two propositions cover all negative measures without a global finiteness assumption; the all-lower branch still carries no moment condition.

The same stationary localization proves necessity at the negative tropical endpoint: convexity is replaced by quasiconvexity, so one first imposes exact stationarity and then uses the negative second derivative. The result is exactly the two alternatives of Proposition 1.5, with the right-hand side of the signed moment bound equal to 0. At the positive tropical endpoint, strong max-quasiconcavity on simplex faces forces τ to be concentrated at $\alpha = 0$: strong max-quasiconcavity would force the integrated entropy to be constant on each fixed-support face, whereas comparing a nonuniform point with the uniform point gives a strict H_α gap for every $\alpha > 0$. Thus the negative tropical endpoint has exactly the measures of Proposition 1.5, whereas the positive tropical endpoint is only $H_{+\infty,0}$ in Eq. (125) of Ref. [4].

2.2 Necessary conditions for derivations

Next, we turn to the necessary conditions for the extremal derivations. According to Eq. (91) and Proposition 3.7 of Ref. [4], these are

$$H_{0,\alpha}(P_{XY}) = \sum_{y\in\mathcal{Y}} P_Y(y) H_\alpha(P_{X|Y=y}). \tag{36}$$

It is therefore sufficient to consider the concavity of

$$\vec{x} \longmapsto \|\vec{x}\|_1 H_\alpha(\hat{x}), \tag{37}$$

as required by Lemma A.2 of Ref. [4]. We now show that every $\alpha > 1$, including $+\infty$, is excluded.

Proposition 2.7. *The function*

$$\vec{x} \longmapsto \|\vec{x}\|_1 H_\alpha(\hat{x}) \tag{38}$$

is concave only if $\alpha \in [0, 1]$.

Proof idea. Use the two-block point and direction from Proposition 2.1. For $\alpha > 1$, put

$$\beta = \frac{\alpha}{1-\alpha} < 0. \tag{39}$$

After the large-dimensional localization, the second derivative at the origin tends, up to a positive factor, to

$$-\frac{\beta}{p^2} > 0. \tag{40}$$

This is a convex direction and contradicts the required concavity. The endpoint $\alpha = +\infty$ follows from the same localization by taking the endpoint limit. □

Importing the upstream representation of derivations from Proposition 3.7 and Eq. (91) of Ref. [4], and combining it with Proposition 1.8 and the semiring lift, gives the final statement.

Proposition 2.8. *The set of derivations of the conditional majorization semiring S at $\|\cdot\|$ is spanned by the quantities $H_{0,\alpha}$ of Definition 6.1 in Ref. [4], with $\alpha \in [0,1]$.*

The span assertion in Proposition 2.8 is imported from that upstream representation. The sufficiency and curvature arguments in this note supply the exact restriction $\alpha \in [0,1]$ for its extremal candidates.

Together, the temperate, tropical, and derivation statements give the exact parameter ranges for the concrete candidate families in Eqs. (122)–(125) of Ref. [4]; see also Definitions 6.2 and 6.3 there for the parameter sets. They supply the parameter-classification input used in Theorem 7.1 of that paper. The upstream representation theorem and the later operational applications remain results of the cited paper rather than claims of this standalone note.

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